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CO-3 AT-1

23.Text Simplifier for Accessibility Agovernment site simplifies circulars to plain language. Task: Using the Gemini API, rewrite an official circular at an 8th-grade reading level while keeping all facts.Demonstrate on two circulars and verify no dates/amounts changed

Code :

```
import os
import re
from google import genai
from dotenv import load_dotenv

load_dotenv()

#
=====
===
# 1. GEMINI API SETUP
#
=====
===

api_key = os.getenv("GEMINI_API_KEY")

if not api_key:
    raise ValueError(
        "GEMINI_API_KEY is missing. "
        "Set your Gemini API key in the .env file."
    )

client = genai.Client(api_key=api_key)

MODEL_NAME = "gemini-2.0-flash"

#
=====
===
# 2. FUNCTION TO EXTRACT DATES AND AMOUNTS
#
=====
===
```

```

def extract_dates_amounts(text):
    patterns = [
        r'\b\d{1,2}[/-]\d{1,2}[/-]\d{2,4}\b',

        r'\b\d{1,2}\s+(?:January|February|March|April|May|June|July|August|September|October|November|December)\s+\d{4}\b',

        r'\b(?:January|February|March|April|May|June|July|August|September|October|November|December)\s+\d{1,2},\s+\d{4}\b',

        r'Rs\.\?s?\d+(?:,\d+)*(?:\.\d+)?',
        r'₹\s?\d+(?:,\d+)*(?:\.\d+)?'
    ]

    results = []

    for pattern in patterns:
        results.extend(re.findall(pattern, text, re.IGNORECASE))

    return results

```

```

#
=====
===
# 3. FUNCTION TO SIMPLIFY CIRCULAR
#
=====
===

```

```
def simplify_circular(circular):
```

```

    prompt = f"""
    You are an accessibility editor.

```

Rewrite the official circular below so that it can be easily understood by students at an 8th-grade reading level.

IMPORTANT RULES:

1. Keep EVERY fact from the original circular.
2. Do NOT add new information.
3. Do NOT remove any information.
4. Do NOT change dates.
5. Do NOT change amounts or fees.
6. Do NOT change names, deadlines, locations, or requirements.
7. Use simple and clear English.
8. Keep the meaning exactly the same.
9. Return only the simplified circular.

ORIGINAL CIRCULAR:

```
{circular}
"""
```

```
    try:
```

```

        response = client.models.generate_content(
            model=MODEL_NAME,
            contents=prompt
        )

        return response.text.strip()

    except Exception as e:
        print("Error: Gemini API request failed.")
        print("Details:", e)
        return None

#
=====
===
# 4. CIRCULAR 1
#
=====
===

circular1 = """
OFFICIAL CIRCULAR 1

All students are hereby informed that the college
scholarship
application must be submitted on or before 15 September
2026.
Eligible students are required to submit the completed
application
form along with a copy of their identity card and income
certificate.
The scholarship processing fee is Rs. 250. Applications
received
after 15 September 2026 will not be accepted.
"""

#
=====
===
# 5. CIRCULAR 2
#
=====
===

circular2 = """
OFFICIAL CIRCULAR 2

Students are informed that the university examination fee
of
Rs. 1,500 must be paid on or before 30 September 2026.
Students must complete the examination registration form
before
making the payment. Late applications submitted after
30 September 2026 will not be considered.
"""

```

```

#
=====
===
# 6. PROCESS AND VERIFY
#
=====
===

def process_circular(number, circular):

    print("\n" + "=" * 60)
    print(f"CIRCULAR {number}")
    print("=" * 60)

    print("\nOriginal Circular:")
    print(circular)

    original_values = extract_dates_amounts(circular)

    simplified = simplify_circular(circular)

    if simplified is None:
        return

    print("\nSimplified Circular:")
    print(simplified)

    simplified_values = extract_dates_amounts(simplified)

    print("\nOriginal dates/amounts:")
    print(original_values)

    print("\nSimplified dates/amounts:")
    print(simplified_values)

    # Verify that every original date/amount is present
    verification = all(
        value.lower() in simplified.lower()
        for value in original_values
    )

    print("\nVerification:")

    if verification:
        print("PASS - No dates or amounts were changed.")
    else:
        print("FAIL - A date or amount may have
changed.")

#
=====
===
# 7. MAIN PROGRAM
#
=====
===

```

```

print("=" * 60)
print("TEXT SIMPLIFIER FOR ACCESSIBILITY")
print("=" * 60)

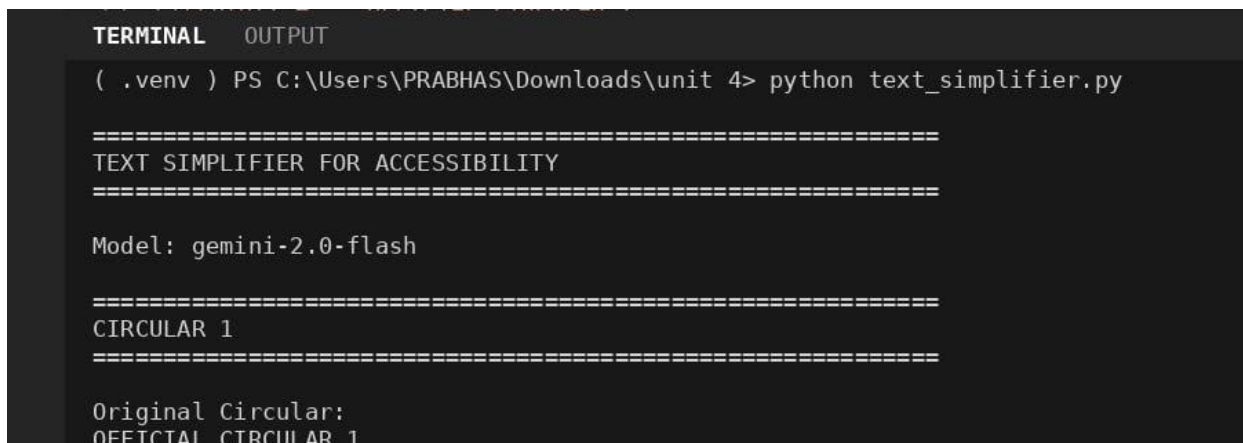
print("\nModel:", MODEL_NAME)

process_circular(1, circular1)
process_circular(2, circular2)

print("\n" + "=" * 60)
print("PROGRAM COMPLETED")
print("=" * 60)

```

Output – VS Code Terminal :



```

TERMINAL  OUTPUT
( .venv ) PS C:\Users\PRABHAS\Downloads\unit 4> python text_simplifier.py

=====
TEXT SIMPLIFIER FOR ACCESSIBILITY
=====

Model: gemini-2.0-flash

=====
CIRCULAR 1
=====

Original Circular:
OFFICIAL CIRCULAR 1

```